ReAct Agent :

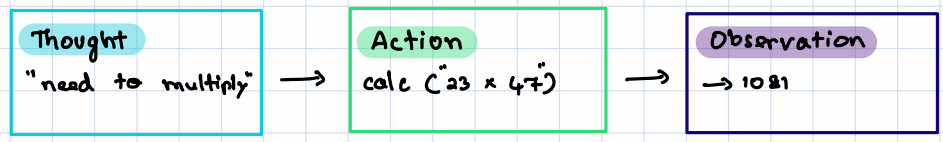
What and Why?

- LLMs reason well, but can't act
- ReAct = Reasoning + Action
- Tools : hands ; (search , db tool , weather , API)

ReAct Agent - Flow:

① User asks a question
↓
② Build a ReAct prompt (tools list + Plan + op format)
↓
③ LLM writes a thought + action (as plain text)
↓
④ Backend code parse the text & call the tool
↓
⑤ Tool returns a result
↓
⑥ Result added to chat-history + loop back to LLM
↓
repeat until LLM writes Final Answer

The 3 - Stage Loop:



repeat until Final Answer

Simple Example:

Q : What is 23 x 47 ?
Thought : need to multiply. So use "calc" tool
Action : calc ("23 x 47")
Observation : 1081
Thought : got the final answer to user query
Final Answer : 23 x 47 is 1081

Note: the LLM only writes the action as a text.
our code does the actual tool call and return the result to the LLM.

When to use ReAct:

✓ Good fit :

- multi-step research
- need external data / real-time data
- tasks with branching logic

✗ overkill :

- simple chat / Q & A
- pure creative writing
- simple summarization
- strict latency budget
